SANTHOSHKUMAR R

Computer Vision & AI/ML Engineer | YOLO Object Detection, Face Embedding Search, AWS SageMaker

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PROFESSIONAL SUMMARY

Cloud, MLOps & Computer Vision Engineer with 4.2+ years of experience building and deploying production ML systems across edge and cloud. Led development of an end-to-end AI-powered CCTV surveillance platform: fine-tuned custom YOLO object detection models (mAP@0.5 improved from ~82% to ~91% on 10k annotated images), built a real-time face recognition system using ArcFace embeddings with pgvector similarity search, and deployed real-time inference on both NVIDIA Jetson Orin Nano edge devices and AWS SageMaker endpoints. Strong hands-on background across the full ML lifecycle — data annotation, model training/fine-tuning, containerized deployment, and CI/CD-driven MLOps — combined with AWS cloud security and compliance expertise (HIPAA, SOC2). AWS Certified Security – Specialty.

TECHNICAL SKILLS

- **Computer Vision & Object Detection:** Custom YOLO model fine-tuning, multi-class object & attribute detection, OpenCV preprocessing, CVAT & Label Studio (dataset annotation), false-positive reduction under variable lighting
- **Face & Embedding Search:** InsightFace / ArcFace facial embeddings, PostgreSQL + pgvector (HNSW indexing), nearest-neighbor similarity search, vision-language model evaluation for semantic image search
- **Edge AI & GPU Infrastructure:** NVIDIA Jetson Orin Nano (edge inference), GStreamer hardware decode/encode (nvv4l2decoder/nvv4l2h264enc), CUDA-accelerated pipelines, dual-GPU local server (2× RTX 5090) for training, LoRA fine-tuning & benchmarking
- **AWS / Cloud ML Services:** SageMaker (Training Jobs & Real-Time Inference Endpoints), MLflow, Lambda, S3, ECR, Step Functions, EC2, VPC, WAF, GuardDuty, Inspector, Config
- **MLOps & Deployment:** Docker, Kubernetes, ECS (Fargate), EKS, Terraform (Infrastructure as Code), containerized inference services
- **Programming:** Python (Pandas, NumPy, Scikit-learn, TensorFlow), SQL, JavaScript, Bash
- **CI/CD & Pipeline Automation:** AWS CodePipeline, CloudWatch/SNS monitoring, Jenkins, Git, Bitbucket, automated retraining & model promotion workflows
- **Security & Compliance:** HIPAA & SOC2 compliance automation, IAM least-privilege, encryption, CloudTrail, zero-trust access

PROFESSIONAL EXPERIENCE

Computer Vision & AI/ML Engineer — Clustrex Data Private Limited 06/2022 – Present

Chennai, India

- Built and fine-tuned custom YOLO object detection models for an AI-powered CCTV surveillance platform across office and sports facilities — detecting people, employee/visitor status, PPE, carried objects, and facility state. improved mAP@0.5 from ~82% to ~91% by fine-tuning on 10k annotated images (CVAT, Label Studio) with targeted augmentation to reduce false positives under variable lighting.
- Achieved real-time inference at 35–45 FPS on NVIDIA Jetson Orin Nano Super edge devices and 100+ FPS on RTX 5090 GPUs, using GStreamer hardware decode/encode and CUDA-accelerated video pipelines for continuous 24/7, multi-camera monitoring.
- Designed a face recognition and identity-search system using InsightFace/ArcFace embeddings stored in PostgreSQL with pgvector (HNSW indexing), enabling real-time nearest-neighbor similarity search for employee identification.
- Deployed production computer vision models as containerized real-time inference endpoints on Amazon SageMaker (ml.g5.xlarge / ml.g5.2xlarge), building Docker inference containers and integrating deployments into CI/CD pipelines with endpoint health monitoring.
- Ran model training and experimentation across hybrid infrastructure — SageMaker Training Jobs for scalable cloud training, and a local dual-GPU server (2× NVIDIA RTX 5090, 64GB RAM) for YOLO training, LoRA fine-tuning of vision-language models, and rapid benchmarking before cloud deployment.
- Extended the platform with LLM-based CCTV summarization and RAG-based search over surveillance summaries, and designed secure AWS infrastructure (VPC, WAF, GuardDuty, Inspector, Config) that raised the cloud security score to 80% while achieving HIPAA & SOC2 compliance.
- Reduced operational costs by 40% and improved system performance by 20% through infrastructure automation and auto-scaling; automated CI/CD pipelines (CodePipeline, ECR) for model artifact management and promotion across development, staging, and production.

Data Science Intern — Vani Analytics 06/2021 – 11/2021

Chennai, India

- Conducted exploratory data analysis and applied statistical/ML techniques (linear/logistic regression, k-means, hierarchical clustering) on real-world healthcare datasets to support model development and evaluation.
- Built interactive visualizations and dashboards using ggplot2 to communicate model insights and data quality findings to stakeholders.

ADDITIONAL PROJECTS

AWS IaaS Automation using Terraform

- Built Terraform-based multi-tenant AWS infrastructure (VPC, EC2, ECS Fargate, Aurora RDS, S3) to support scalable, secure ML and application deployments.
- Integrated serverless and networking services — Lambda, API Gateway (JWT authorizers), WAF, Route53, ACM SSL, CodePipeline, CloudWatch/SNS — for production-grade, monitored cloud infrastructure.

HIPAA & SOC2 Compliance Automation — AWS

- Automated AWS infrastructure with Terraform embedding HIPAA & SOC2 compliance controls, including encryption, IAM least-privilege, VPC segmentation, CloudWatch monitoring, and CloudTrail logging.
- Secured EC2 access using Systems Manager (SSM) and SSH tunneling via ProxyCommand, eliminating public IPs, open SSH ports, and bastion hosts; enforced zero-trust access with IAM + MFA.

EDUCATION

Master of Science in Artificial Intelligence — Sri Ramachandra Institute of Higher Education and Research 08/2024

- Focus on optimization algorithms for ML models, deep learning, and neural networks; completed part-time while working full-time at Clustrex.

Bachelor of Computer Applications — Guru Nanak College, University of Madras 07/2022

- Minors in Mathematics and Computational Algorithms.

CERTIFICATIONS

- AWS Certified Security – Specialty (SCS-C03)
- AWS Certified Generative AI Developer Professional (AIP-C01) — In Progress
- HIPAA Awareness for Business Associates Certificate
- Data Science Using R